Response to Reviewers

"On the Monotonicity of Information Measures"

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Working-template note. The opening summary reflects the October 28 draft. The individual responses and planning notes below remain work in progress. Before finalizing the letter, replace the remaining future-tense statements and placeholders with factual accounts and verified manuscript references. [Todo: *Delete this note and all drafting comments, and remove revision colors, after the responses are complete.*]

Dear Professor Ottaviani,

Thank you for the opportunity to revise our manuscript. We are grateful to you, the Associate Editor, and both referees for the careful and constructive reports. Following your guidance, we have substantially revised the manuscript to make the contribution, the source of the technical difficulty, and the economic significance of the results clear to a broad audience of economic theorists.

The main changes in the revised manuscript are as follows:

1. **Reframing the scope.** We have reframed the paper to study the monotonicity of information measures, functions that assign a numerical value to each statistical experiment. This framing encompasses our previous focus on information costs, as well as additionally statistical divergences, the value of information, and preferences for early or late resolution of uncertainty.

2. **Clarifying the main Lehmann characterization.** We have made the Lehmann characterization in Theorem 2 the central focus of the paper and substantially revised its exposition. Section 2.1 now introduces the Lehmann order through Kim's (2023) monotone quasi-garbling characterization, in parallel with the garbling definition of the Blackwell order, making their relationship more transparent. Section 4.2 now isolates the path construction in Proposition 1, a structural result about the Lehmann order that yields sufficiency in Theorem 2. A detailed, step-by-step proof sketch explains this construction with graphical and geometric illustrations. We have also improved the readability of the appendix proof and formalized steps that were informal in the previous submission.

3. **Adding a posterior-space characterization.** We have added Section 5 to formalize information measures defined on random posteriors and translate our first-order conditions

into this setting. The resulting conditions apply directly to information measures in posterior space, which also connects to the local-utility characterization of Blackwell monotonicity in Grant, Kajii, and Polak (1998).

4. **Characterizing the Lehmann order.** As a new application of our results, we characterize the Lehmann order through a class of posterior-separable information measures. This extends the use of our characterizations beyond checking the monotonicity of a given measure.

5. **Streamlining the exposition.** Following the suggestions, we have removed the standalone binary-experiment section and streamlined the writing throughout. We have also moved the quasi-convexity result from the online appendix to the main text.

In addition to these major changes, we have made a number of improvements in response to the detailed feedback. We hope the revised version addresses all of your comments and concerns. Below we provide a point-by-point response to each comment.

Sincerely,

Xiaoyu Cheng and Yonggyun Kim

Response to the Editor

> Building on the AE's comments, I would encourage you not only to clarify what is difficult or technically demanding in your exercise, but also to find a way to explain the main results in a simple and intuitive manner. Although a substantial part of the feedback concerns presentation, the revision we are inviting is a major one. In particular, considerable work will be needed to sharpen the framing of the contribution and to make the paper accessible and appealing to a broad audience of theorists, beyond researchers working specifically on the comparison of experiments.
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> I would especially like to emphasize a concern raised by the reviewers about the technical style of the paper. While I personally appreciate work on the Lehmann order, the reviewers' broader comments reinforce my concern that the current draft is written more in the style of a highly technical journal, such as the Journal of Mathematical Economics, than for a general-interest theory journal such as JET.
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> I fully appreciate that it is not always easy to distill technical results and make their significance transparent to a general theory audience. Nevertheless, a key condition for publication will be to present your results in a substantially clearer, more intuitive, and more engaging way. I personally find graphical and geometric representations particularly effective in this respect, so I would encourage you to develop this aspect further where useful. I am not explicitly asking for applications, but any effort to illustrate the economic content and significance of your results would help considerably in motivating the paper, engaging readers, and making the contribution more broadly appealing.

Response. Thank you for the constructive guidance for the revision. We have approached the revision by reconsidering the paper's framing and exposition as a whole. Specifically, we start by framing the paper around information measures, which brings together several familiar objects in the literature, including information costs, values of information, and statistical divergences, highlighting their common concern with how a measure responds to more information. We then organize the paper around a simple central insight: global monotonicity can be characterized by first-order conditions derived from local depreciation of information.

For exposition, our strategy is to develop this insight intuitively before introducing the technical argument. To this end, we now define Lehmann order using Kim (2023)'s monotone quasi-garbling characterization, which parallels the garbling definition of the Blackwell order. This makes what Lehmann monotonicity requires relative to Blackwell monotonicity

more transparent. We have also substantially improved the exposition of the Lehmann characterization, using graphical and geometric illustrations to explain the construction, and make the purpose of each step clear. This progression is intended to help readers understand both why the result is useful and where the difficulty lies. We have also streamlined the organization so that the main text presents a coherent argument, supported by detailed proofs in the appendices.

To communicate the broader significance, we connect the analysis to the familiar language of posterior beliefs and expand the discussion of what the results enable researchers to do. The revised paper shows how the conditions apply to widely used information measures and how they yield further insights into information comparisons. For researchers studying information acquisition, information design, or the value of information, our characterizations provide a tractable way to determine whether a given cost or value increases with informativeness by examining local changes at individual experiments. We hope this makes the paper useful as a tool for assessing information measures and developing comparative statics with respect to information, including for theorists who do not specialize in the comparison of experiments.

Response to the Associate Editor

> I have now heard back from two expert reviewers whose advice I sought. Both referees are overall positive, although they raise important concerns. Like the referees, I think the paper contains a number of interesting insights. The reviews are broadly in line with one another, but let me summarize what I view as the main points to tackle:

Response. Thank you very much for your careful reading of our paper and for bringing together the referees' concerns. Your guidance helped us see more clearly where the exposition needed to be improved and how to explain the contribution and the technical difficulty.

Comment 1

> The authors should explain more precisely what is new and where the technical innovation lies. The paper describes the first general characterization of Lehmann monotonicity as its main contribution and identifies the construction of informativeness-reducing paths as its principal technical contribution. What remains unclear is how these results differ from, and go beyond, the existing literature on the Lehmann order.
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> Referee 1 also asks the authors to explain whether, and to what extent, the characterizations can be reformulated in terms of posterior beliefs.
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> Referee 2 similarly asks for a clearer account of what is new relative to the existing characterizations of the Lehmann order. These comments should not be handled as a simple checklist but as signs that the paper is not well structured. The authors should reconsider the exposition as a whole, so that the contribution and the source of the technical difficulty are transparent.

Response. We have revised the Introduction, including the Related Literature, and Section 4, added the posterior-space formulation in Section 5, and reorganized the proof of the Lehmann characterization. We agree that our earlier exposition did not explain clearly enough how the results build on the existing literature or where the technical difficulty lies. Your comment helped us see the need to reconsider the exposition as a whole.

First, we now state more directly how our results differ from existing work on the Lehmann order. Existing characterizations (Jewitt, 2007; Kim, 1995; Chi, 2015; Asseyer, 2025) describe which experiments the order ranks, rather than which information measures respect the ranking. Verifying Lehmann monotonicity through them therefore still requires comparing every pair of ranked experiments. Theorem 2 instead reduces this verification to first-order conditions at each experiment. The Introduction and Related Literature now make this

distinction explicit.

Second, Section 2.1 now introduces the Lehmann order through the equivalent monotone quasi-garbling (Kim, 2023), in parallel with the garbling definition of the Blackwell order. Section 4.2.1 then explains how we use the characterizations of Kim (2023) and Jewitt (2007). Kim's monotone quasi-garbling representation motivates the necessary local inequalities, while Jewitt's geometric characterization identifies the target through comparisons of probability-probability (PP) plots. These results provide valuable starting points, but do not by themselves show how to move to the target through local changes, which is needed to establish sufficiency of the local inequalities.

To establish sufficiency, we prove a local-generation result for the Lehmann order (Proposition 1): every Lehmann comparison between positive experiments is generated, up to splitting and merging signals, by reverse signal replacements along paths that remain within the MLRP set. This is a structural property of the Lehmann order itself; once it is established, sufficiency of the local inequalities follows from the chain rule. Section 4.2.2 explains the two obstacles to constructing such paths: the MLRP set is nonconvex, so straight-line interpolation can leave the domain; and the adjacent-state PP plots are interdependent, so changing one can disturb the others. Our key building block is chord replacement (Lemma B.3), which raises one selected PP plot toward its target while preserving the MLRP and leaving all other adjacent-state PP plots unchanged. Repeating this operation gives the required global path.

Third, we have added Section 5 in response to the request for a posterior formulation. A direct chain-rule argument translates our conditions from likelihood space into posterior coordinates. At a fixed full-support prior, Theorem 3 gives necessary and sufficient first-order conditions for Blackwell and Lehmann monotonicity of continuous and differentiable information measures on finite-support distributions of posterior beliefs.

Fourth, we have reorganized the proof to help readers follow the main argument before turning to the technical details. Section 4.2 explains the three steps—PP geometry, chord replacement, and the construction of a global path—with geometric illustrations in Figures 1 and 2. Appendix B.4 follows the same structure, proving Proposition 1 and working through the two-column and three-column constructions of the chord-replacement lemma (Lemma B.3), while the general induction and proofs of supporting technical lemmas appear in Online Appendix OA.1. We have also formalized steps that were informal in the previous submission and removed the standalone binary-experiment section. We hope these changes make the general Lehmann argument easier to follow.

Comment 2

> As Referee 1 points out, although the paper is motivated by information costs, the results apply more generally to functionals over experiments. It is true that information costs may be the most active application of functionals over experiments. Nevertheless, the Blackwell order and the convex order have a long history in economic theory and operations research, and functionals over experiments appear in a variety of settings. To my reading, there is no reason why the results should be framed as a study of information costs.

Response. Thank you for this suggestion. We have reframed the paper around *information measures*, functions that assign a numerical value to each finite statistical experiment, and retitled it "On the Monotonicity of Information Measures." The Introduction and Section 2 now present this framing, which encompasses information costs as well as statistical divergences, the value of information, and preferences for early or late resolution of uncertainty. Information costs remain a leading source of examples, but the theorem statements and their discussion make clear that the results do not depend on a cost interpretation. We discuss the connection to the broader literature on the Blackwell order and the convex order in our response to Comment 3.

Comment 3

> The authors should engage much more extensively with the older literature on the comparison of statistical experiments and with the body of work on convex order, majorization, and stochastic matrices. I was surprised to see that most references are economics papers from the last twenty years.
>
> I am not asking for an encyclopedic survey. However, for a paper whose contribution is primarily technical, the authors should explain carefully how their results relate to the existing theory. In particular, the authors should give the reader a clear basis for assessing which parts of the results and proofs are genuinely new. Ideally, the paper should be written so that a reader from OR or information theory could appreciate it with little effort.

Response. Thank you for this comment. We have expanded the Related Literature to relate our results to the classical theory.

The comparison of statistical experiments has a long history in statistics, beginning with Blackwell (1951, 1953) and developed by Le Cam (1964) and Torgersen (1991), who also study functionals of experiments and their connection to majorization. Majorization is the

Blackwell order restricted to two-state experiments in which one state's signal distribution is uniform, and the Schur–Ostrowski criterion characterizes the monotone (Schur-convex) functions by symmetry together with a first-order condition (Marshall, Olkin, and Arnold, 2011). Theorem 1 extends this type of characterization to experiments with any finite number of states and signals, and Theorem 2 provides its counterpart for the Lehmann order.

In the literature on stochastic orders, orders are often characterized by the classes of functions whose expectations they preserve (Shaked and Shanthikumar, 2007); the classical instance is the equivalence between the Blackwell order and the convex order on posterior beliefs (Blackwell, 1953; Strassen, 1965; Cartier, Fell, and Meyer, 1964). Section 6.2 provides an analogous characterization of the Lehmann order through posterior-mean-ratio measures, a class that includes every f-divergence between the signal distributions of two states.

We also clarify the relation to existing characterizations of the Lehmann order (Jewitt, 2007; Kim, 1995; Chi, 2015; Asseyer, 2025), which describe which experiments are ranked rather than which measures respect the ranking, and to the economics literature on Blackwell monotonicity, where the results for separable measures and the local-utility criterion of Grant, Kajii, and Polak (1998) were known and are recovered by our characterization.

Comment 4

> The case where the functional C is convex or quasi-convex should have more prominence, since these are natural assumptions.

Response. Thank you for this suggestion. We have moved the quasi-convexity result from the online appendix to Section 6.3 of the main text (Theorem 5), together with a sketch of its proof. Since quasi-convexity is weaker than convexity, the result also covers the convex case commonly assumed in applications, as we note in the text.

The main benefit of quasi-convexity here is that it allows us to obtain a first-order characterization of Blackwell monotonicity when the information measure is defined only on a fixed signal space. We explored whether quasi-convexity could give a similar characterization for Lehmann monotonicity. However, the nonconvexity of the MLRP set makes an analogous argument challenging. We discuss this difficulty at the end of Section 6.3 and leave a fixed-signal-space characterization of Lehmann monotonicity under quasi-convexity for future research.

Comment 5

> Like Referee 1, I don't think it is ideal to have separate sections for the binary and the general case. Experiments with binary states can be motivated as simple-hypothesis tests, but binary-state-binary-signals are very special. Is there a way to fold the binary case in the general case?

Response. Thank you for this suggestion. We have removed the standalone binary-experiment section and now focus on the general characterization results. We were able to explain and illustrate both Blackwell and Lehmann monotonicity in the general case without referring back to the binary case. Introducing the Lehmann order through monotone quasi-garbling in Section 2.1 gives a direct parallel with the Blackwell garbling definition and helps provide intuition without a separate binary treatment. Only in the Introduction we use a two-state, two-signal experiment to illustrate signal replacements and reverse signal replacements as special cases of the general operations.

Additional comments

Additional comment 1

> The notation f for an experiment is highly non-standard for what is just a stochastic matrix. The traditional notation for an experiment is P, with $P_\theta(s)$ the probability of state s given state θ. Alternatively, the standard notation for a stochastic matrix is P or Π, with entries P_{ij} or π_{ij}.

Response. Thank you for this suggestion. We have changed the notation for an experiment from f to P throughout the paper.

Additional comment 2

> Since the domain of C is a subset of a vector space that has empty interior, the standard textbook definition of differentiability does not apply. Some care is needed.

Response. Thank you for pointing this out. In Section 2.2, we have clarified that differentiability of an information measure $\mathcal{I}_m$, defined on the set of m-signal experiments, is required on the relative interior of its domain and is understood relative to its affine hull. Its gradient then represents directional derivatives in feasible directions as $\langle \nabla \mathcal{I}_m(P), Q - P \rangle$.

Additional comment 3

> Like Referee 1, I found some of the proofs hard to follow.

Response. Thank you for pointing this out. We have revised the proof sketch of Theorem 2 in Section 4.2 to explain the construction step by step with geometric illustrations. Appendix B.4 follows the same roadmap and develops the key constructions, while the general induction and proofs of supporting lemmas appear in Online Appendix OA.1. We have also supplied formal arguments for steps left informal in the previous submission. Our response to Referee 1's Comment 4 describes these revisions in more detail.

More generally, we have simplified the presentation of the other proofs throughout the paper to make the reasoning easier to follow.

Response to Referee 1

The paper provides new characterizations of monotonicity, with respect to both the Blackwell and Lehmann orders, for functionals defined on finite-signal experiments. The characterizations are expressed as first-order conditions with respect to signal likelihoods. The main idea is to construct information-degrading paths connecting a given experiment to a dominated one by shifting probability mass from one signal to another, in a way that reflects the relevant garbling operation.

I found the approach interesting and, in hindsight, very natural. At the same time, I have some reservations about the practical usefulness of the characterizations as verification tools. For Blackwell monotonicity, other known criteria often seem easier to apply in standard classes of costs, as the authors themselves acknowledge. For Lehmann monotonicity, the characterization is more novel, but it is not always clear how tractable the reverse-signal-replacement conditions are in general, although they are useful in some of the examples considered in the paper. That said, I do not think that the applicability of the characterization needs to be fully settled for the paper to make a contribution. A structural characterization can be valuable even when its direct applicability varies across classes of functionals.

Overall, I think the paper would be substantially improved by several revisions and clarifications.

Response. Thank you very much for your careful reading of our paper and for recognizing the value of a structural characterization. We are especially grateful for your constructive suggestions to formulate our conditions in posterior space and connect them to Grant, Kajii, and Polak (1998). These suggestions led us to add Section 5 and helped us clarify how our results relate to their characterization. Your comments also helped us improve the exposition of the proofs and formalize steps that were previously informal.

Comment 1

It would be useful to provide equivalent first-order conditions in terms of induced posteriors and signal probabilities. Many readers, especially in the rational inattention and preferences-over-information literatures, work directly with random posteriors rather than likelihood matrices. Thus, when a functional can be written as a functional of the induced distribution of posteriors (when such a posterior representation is available, as is natural under Blackwell monotonicity and in many applications), it would be helpful

to rewrite the paper's conditions in terms of posterior beliefs q^j and signal probabilities τ_j.

Section 4.2 partially moves in this direction by applying the characterization to posterior-separable costs. I would encourage the authors to first discuss more general functionals of random posteriors and then specialize to posterior-separable costs. This would make the characterization more accessible and would clarify how the likelihood-based conditions translate into the language commonly used in applications.

Response. Following your suggestion, we have added Section 5 which develops the characterization for general information measures defined on distributions of posteriors. We show that our first-order conditions from likelihood space can be translated into posterior coordinates through a direct chain-rule argument. Theorem 3 gives the resulting necessary and sufficient conditions for Blackwell and Lehmann monotonicity of continuous and differentiable measures in this setting. The conditions have a marginal-value interpretation: for each signal replacement or reverse signal replacement, the destination signal must assign lower marginal value to the transferred probability mass than the source signal.

We have retained the discussion of posterior-separable measures in Section 6.1, where we now present it as a direct application of the general characterization, as you suggested.

This reformulation also inspired us to explore whether such information measures could characterize the Lehmann order itself. Section 6.2 gives such a characterization for positive MLRP experiments through posterior-mean-ratio measures (Theorem 4).

Comment 2

Although the paper is motivated by information costs, the results apply more generally to functionals over experiments. I think this point should be made more explicit. In particular, the Blackwell-monotonicity characterization can be related to preferences for earlier resolution of uncertainty in recursive utility models. Analogously, reversing the inequality in the first-order condition would correspond to a preference for later resolution. Pointing out this connection, while specifying the relation of their condition to the convexity of Gâteaux derivative in Grant, Kajii, and Polak (1998), would broaden and clarify the perceived scope of the contribution.

Response. Than you for directing us to Grant, Kajii, and Polak (1998, GKP), which provides a valuable connection that we now discuss in Section 6.1.2.

First, we have reframed the paper to focus on general information measures, as explained

in the Introduction and Section 2. This framing encompasses information costs as well as preferences for early or late resolution of uncertainty.

Second, under additional regularity conditions, our first-order criterion for Blackwell monotonicity is equivalent to convexity of each local utility, recovering GKP's characterization in our setting. However, it is important to note that, given an information measure, differentiability in posterior coordinates and the existence of local utilities do not imply one another. Therefore, the equivalence with GKP is valid only for information measures satisfy both sets of conditions simultaneously.

Within this common domain, applying the same translation to our Lehmann condition yields a parallel local-utility characterization of Lehmann monotonicity, which provides an extension of the GKP characterization to the case of Lehmann monotonicity.

Comment 3

> I am not fully convinced that the separate treatment of the binary-signal case, while useful for intuition, is necessary in its current form, given that the paper is already quite long. Moreover, repeatedly referring back to definitions in the binary case while explaining the general finite-signal case sometimes makes the latter harder to follow. I would consider removing the binary case and try to illustrate the general case as clearly as possible, perhaps only retaining the figures and the main intuition of the binary case to help such explanation.

Response. Thank you for this suggestion. As explained in our response to the Associate Editor's Comment 5, we have removed the standalone binary-experiment section. Section 2.1 now introduces the Lehmann order through monotone quasi-garbling alongside the Blackwell garbling definition, providing a common starting point for the general analysis. Sections 3 and 4 now present and illustrate the general Blackwell and Lehmann characterizations directly, without referring to definitions or results for the binary case.

Comment 4

> I found the proof of the finite-signal Lehmann characterization difficult to follow, and at some points too informal for what appears to be one of the central results of the paper. I do not have a precise prescription for how best to reorganize the proof, but I am confident that some improvement in clarity is possible.
>
> As a general suggestion, I would encourage the authors to provide more detailed proof

roadmaps before entering the formal construction, and perhaps also at key intermediate stages of the argument. The notation in this part of the proof is quite demanding. Since only a few indices are active at each step of the construction, it may be possible to simplify the exposition by temporarily suppressing irrelevant indices.

Regarding formality, the induction argument on k in Lemma B.2 should be stated and proved explicitly. Similarly, the perturbation argument used to handle paths should be written out carefully. I understand that including all details in the main text may be cumbersome, but I would recommend providing a precise version in the online appendix and referring to it in the main proof.

Response. We agree that our earlier proof was not easy to follow and left several steps informal. We have therefore revised the exposition of the Lehmann characterization in Section 4.2 and Appendices B.3 and B.4, with supporting formal arguments in Online Appendix OA.1. Thank you for these concrete suggestions, which helped us reorganize the proof and formalize its key steps.

First, we now distinguish the construction in Proposition 1, which concerns the Lehmann order itself, from its use to prove Theorem 2. Appendix B.4 establishes Proposition 1, while Appendix B.3 uses it to prove sufficiency in Theorem 2. We organize the proof sketch of Proposition 1 in Section 4.2 around three steps: the geometry of PP plots, chord replacement, and the construction of a global path through successive chord replacements. Figures 1 and 2 illustrate these steps geometrically. We also use arrows in the displayed matrices to show how the coordinated reverse signal replacements move points in the PP plots. We then explain how chord replacement changes one selected PP plot while preserving the MLRP and leaving the other adjacent-state PP plots unchanged, so that the target plots can be matched successively. We hope this organization helps readers understand the overall logic of the argument before working through its technical details.

Second, we have simplified the notation by suppressing indices that remain fixed within an argument. In Appendix B.4.1, we work with a fixed adjacent state pair and starting column, and omit these indices from the notation for cumulative coordinates and chord slopes.

Third, to improve the readability of the main proof in Appendix B.4.1, we work through the cases $k = 2$ and $k = 3$ of the chord-replacement lemma (Lemma B.3) explicitly. The two-column case establishes the intuition of the basic rotation. The three-column case introduces collisions and explains how the construction continues when moving cumulative points meet, with Figure B.1 illustrating the possibilities. We then preview how the same construction extends as k increases. The general induction follows the same logic as the

three-column case, but involves more collision patterns and notation, so we have placed its full statement and proof in Online Appendix OA.1. We hope the main appendix gives readers a clear sense of how the induction proceeds and why the required properties are preserved. The online appendix provides the complete verification.

Fourth, we have replaced the earlier perturbation step with Lemma B.2, stated in Appendix B.3. For a differentiable information measure that is decreasing in reverse signal replacement, the lemma extends the first-order inequalities from strict to equal likelihood-product cutoffs at positive MLRP experiments. This allows us to apply the inequalities directly along the aligned segments of the coordinated rotation when deducing Theorem 2 from Proposition 1. Following your suggestion, we state the lemma in the main appendix and give the formal proof in the online appendix.

Minor comments

Minor comment 1

> In connection to points 1 and 2 above, when discussing posterior-separable costs/recursive expected utilities, it would be useful to point out as a corollary, and perhaps verify directly, the equivalence between the Blackwell-monotonicity first-order condition and concavity of entropy H/convexity of utility u. This would be subsumed by directly verifying equivalence with convexity of Gâteaux derivative (when admitted).

Response. Corollary 1 in Section 6.1 states that a posterior-separable information measure is Blackwell monotone if and only if its generating function is convex, or equivalently, its entropy function is concave when the measure is written as an expected entropy reduction. The calculation immediately preceding the corollary verifies this directly from our first-order condition. For the more general connection to GKP's local-utility characterization, please see our response to Comment 2, Remark 2 in Section 6.1.2, and Online Appendix OA.2 of the revised paper.

Minor comment 2

> If footnote 5 is kept, it should specify that the objective needs to be linear in probabilities, for example an expected payoff, for binary signals to be without loss.

Response. Thank you for this clarification. We have removed the binary-experiment section, including the former footnote 5.

Minor comment 3

> If Lemma 2 is kept, condition (5) should be expressed in terms of products rather than ratios, in order to deal cleanly with boundary cases.

Response. Thank you for this suggestion. We now use the cross-product formulation, which remains well-defined when likelihoods are zero and thus handles boundary cases.

Minor comment 4

> I believe there is a typo in the proof of Lemma B.2 towards the end of page 34: should it be that $\Phi^{j+2}(t) = F^{j+2}$?

Response. Thank you for this comment. The original sentence has been subsumed by the rewritten proof of the chord-replacement lemma, now Lemma B.3 in Appendix B.4.1.

Minor comment 5

> I believe there is a typo at the end of the proof of Proposition 4: should it be that q^{j+1} FOSD q^j?

Response. Thank you for pointing this out. The original sentence has been subsumed by the new proof of Corollary 2, which now establishes Lehmann monotonicity of mutual information.

Response to Referee 2

> This paper studies different specification of information cost functions which are monotonic with respect to two common orders of information structures. The main results are the characterizations of (1) information costs that are increasing in the Blackwell order, and (2) information costs that are increasing in the Lehmann order.
>
> The results are quite clever, relying largely on the geometric representations of the two orders. It is also appreciated the fact that the analysis is first carried out for binary signal spaces, which allows the reader to visualize the structure of the two orders. Once it is understood how information structures are ordered by each of the two orders, obtaining increasing cost functions is facilitated.
>
> So overall, I see value in this work, but there is still work to be done. One issue has to do with further elaboration on the Lehmann order, which is not as well known as Blackwell. The second issue is that this is primarily a technical paper, and I am not fully convinced that the technical contribution is really that substantial.

Response. Thank you very much for your comments and for recognizing the value of the geometric representations. We have revised the paper to give a more accessible introduction to the Lehmann order and to explain more clearly the contribution of our path construction and its implications. We will response to these two issues in more detail below.

Following the suggestions of the Associate Editor and Referee 1, we have removed the standalone binary-experiment section and its illustrations to make room for the new results and improve the flow of the paper. We have aimed to provide the same level of intuition directly in the general analysis through the following changes:

Section 2.1 now introduces the Lehmann order through Kim's (2023) monotone quasi-garbling characterization, making the parallel with the Blackwell garbling definition more transparent. Sections 3 and 4.1 then derive signal replacements and reverse signal replacements step by step from the respective garbling representations. Finally, the expanded proof sketch in Section 4.2 provides geometric illustrations of the Lehmann construction. We hope this allows readers to understand its main logic without working through the full technical proof.

Comment 1

> The Blackwell order is well understood by most potential readers. In particular, the fact that two information structures are ranked by $\succeq_B$ whenever one is a garbling of the other, allows us to easily see why garbling is geometrically represented by the parallelogram hull (in Figure 1). And once this is established, the two corresponding results (Thms 1 and 3) follow relatively easily. However, the same cannot be said about the Lehmann order, the corresponding representation of which I found difficult to follow and digest. Given the fact that the typical reader will be much less familiar with this order, I would ask the authors to further elaborate on the characterization of $\succeq_L$. Moreover, I would like more extensive discussion on the application of the Lehmann order which are now briefly mentioned in the introduction.

Response. We agree that our previous manuscript does not provide a sufficiently accessible introduction to the Lehmann order. We have revised the paper to address this concern through the following changes:

First, Section 2.1 now introduces the Lehmann order through Kim's (2023) monotone quasi-garbling characterization, alongside the familiar garbling definition of the Blackwell order (Definition 1). Intuitively, a Blackwell garbling applies the same rule for transforming an observed signal in every state. A monotone quasi-garbling allows this rule to vary across states, with the restriction that, conditional on any original signal, the resulting signal is more likely to be high in lower states. Using the same rule in every state satisfies this restriction, so ordinary garbling is a special case.

Sections 3 and 4.1 then develop signal replacements and reverse signal replacements from these representations and explain how they yield the local necessary conditions. In Section 4.2.1, we connect this interpretation to Jewitt's (2007) geometric characterization through PP plots. We explain what the plots represent and how Lehmann dominance corresponds to inclusions between the regions above them. By presenting and discussing both characterizations, we hope to make the order easier to understand through both its garbling interpretation and its geometry, before readers turn to the path construction.

Second, we have expanded the discussion of the Lehmann order's economic relevance in the Introduction and Related Literature. We emphasize in both the Introduction and Section 2.1 that Blackwell dominance requires a weakly higher expected payoff in every decision problem, whereas Lehmann dominance requires it only in monotone decision problems. The latter is tailored to settings where higher actions become more attractive in higher states, such as auctions, screening, and many comparative-statics settings, and therefore permits comparisons

that Blackwell cannot rank.

Additionally, Related Literature now identifies applications in auctions (Persico, 2000; Bobkova, 2024; Frick et al., 2026), finance (Bond, 2023), screening (Kim, 2023; Asseyer, 2025), and grantmaking (Adda and Ottaviani, 2024), as well as extensions to ambiguity-averse decision makers (Li and Zhou, 2020). It also discusses Kim's (1995) information criterion in moral hazard, which is equivalent to the Lehmann order under the MLRP. We hope these references help illustrate the practical relevance of the Lehmann order.

Comment 2

> In each of the four main results, the underlying logic is similar. The authors start with a characterization of the respective order, and then proceed to characterize cost functions that are increasing wrt this order. I find the first step (i.e., the characterization of the orders) less novel, given earlier work in this literature (e.g., Jewitt). This leaves me with the second step, meaning that the paper's main contribution is technical. However, it is not clear how substantial this contribution is. For example, in the Blackwell results the second step seems relatively simple. This leaves me with the path construction in the Lehmann results. I would invite the authors to persuade me that this step is indeed sufficiently rich for JET publication.

Response. We appreciate your insightful summary of the overall structure of our results. As you correctly point out, the existing characterizations of the Lehmann order by Kim (2023) and Jewitt (2007) provide valuable starting points for our analysis. More specifically, as Section 4.2.1 now explains, Kim's monotone quasi-garbling representation includes reverse signal replacements as special cases and thereby motivates the necessary local inequalities, while Jewitt's PP-plot characterization provides a useful tool for comparing experiments under the Lehmann order and identifies the target comparisons. Unlike in the Blackwell case, however, these representations do not by themselves provide a direct argument establishing that the local inequalities suffice for global monotonicity.

Sections 4.2.2–4.2.3 explain the two main obstacles. First, the set of MLRP experiments is non-convex, so straight-line interpolation between ranked experiments can leave the domain. As a result, the simple approach of connecting experiments by straight-line paths in the Blackwell case does not directly apply here. Second, the adjacent-state PP plots are interdependent: changing signal probabilities to move one plot toward its target can disturb the others. Thus, constructing paths separately for each pair of states does not by itself produce a valid path for the whole experiment.

Our main constructive contribution is the local-generation result in Proposition 1: every Lehmann comparison between positive MLRP experiments is generated, up to splitting and merging signals, by reverse signal replacements along paths that remain within the MLRP set. The key building block in our construction is chord replacement (Lemma B.3). It continuously moves the current PP plot toward its target by replacing a portion of that plot with its chord, while preserving the MLRP and leaving all other adjacent-state PP plots unchanged. We also show that chord replacement can be achieved through coordinated reverse signal replacements. A crucial part of the construction is choosing this coordination carefully so that the MLRP is preserved, all other PP plots remain unchanged, and the overall path decreases monotonically in the Lehmann order.

We then show that successive chord replacements can be used to match the target PP plots one at a time without undoing earlier steps. Section 4.2 illustrates this construction, with the proof in Appendix B.4 and the general induction in Online Appendix OA.1. Proposition 1 concerns the Lehmann order itself and does not depend on an information measure. The chain rule then shows that a measure satisfying the local derivative inequalities decreases along these paths; split invariance and continuity yield sufficiency in Theorem 2. We believe that this local-generation characterization of the Lehmann order may also be useful to researchers working with this information order on questions beyond the characterization of monotonicity.

Finally, in our view, while the constructive characterization of the Lehmann order constitutes one of the main contributions of our paper, its implications for the analysis of information measures and the development of practical criteria for assessing monotonicity are equally significant. In particular, the revised paper also develops the implications of this central result within a common likelihood- and posterior-space framework for information measures. In Section 6.1, Propositions 2(b) and 4 provide necessary and sufficient conditions for Lehmann monotonicity of likelihood-separable measures with sublinear generators and of posterior-separable measures at every full-support prior, respectively. These conditions also establish that the widely used entropy cost is Lehmann monotone (Corollary 2). In Section 6.2, we use the characterization to establish Lehmann monotonicity of posterior-mean-ratio measures and show that comparisons using this class characterize the Lehmann order on positive MLRP experiments (Theorem 4). These applications make the first-order characterization a practical criterion for assessing information measures and a tool for deriving further characterizations of the order.

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